

→ consisting of several terms

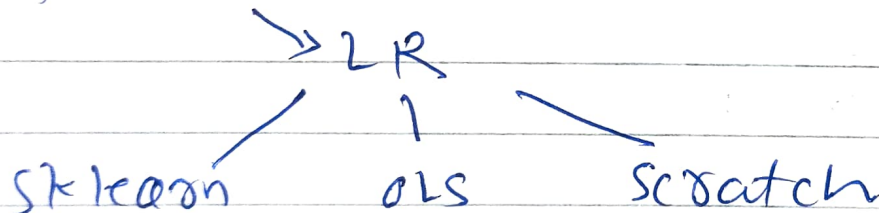
LG: ML: Polynomial Regression

Bias - Variance

Agenda

- Gradient Descent Variants
- Polynomial Regression
- Generalisation & Occam's Razor
- Underfitting & Overfitting
- Bias - Variance Trade off

Revision



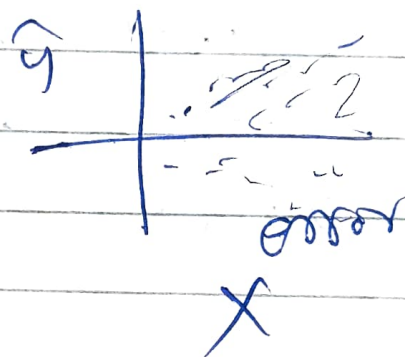
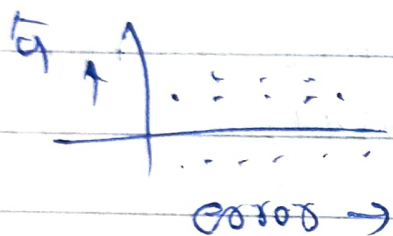
↓
GD

$$w = w_{old} - \eta \frac{dL}{dw}$$

$$L = \text{MSE}$$

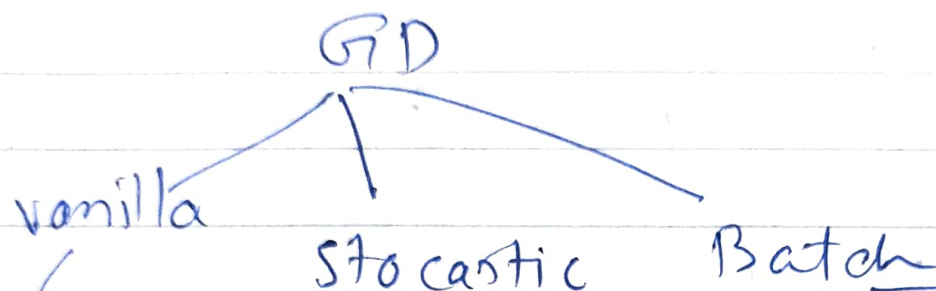
LR assumption

- ① Linear
- ② error should be normally distributed
- ③ error vs $\hat{y}$ should be homoscedasticity not heteroscedasticity



④ $VIF < 5 \rightarrow$ no multi-collinearity

⑤ No auto-correlation



Stochastic
↓
associates with randomness

takes all points

then update w

$$w = w_0 - \eta \frac{dL}{dw}$$

→ instead of all points
K points

$$\text{batches} = N/K$$

$$-\frac{2}{N} \sum (y - \hat{y}) \times (-x_i)$$

time intensive

first take ^{1st} batch
K points update
 w

then take next
batch having K
points then
update w

Each batch have different
point

After finishing all K batch
its called 1 epoch.

Stochastic GD

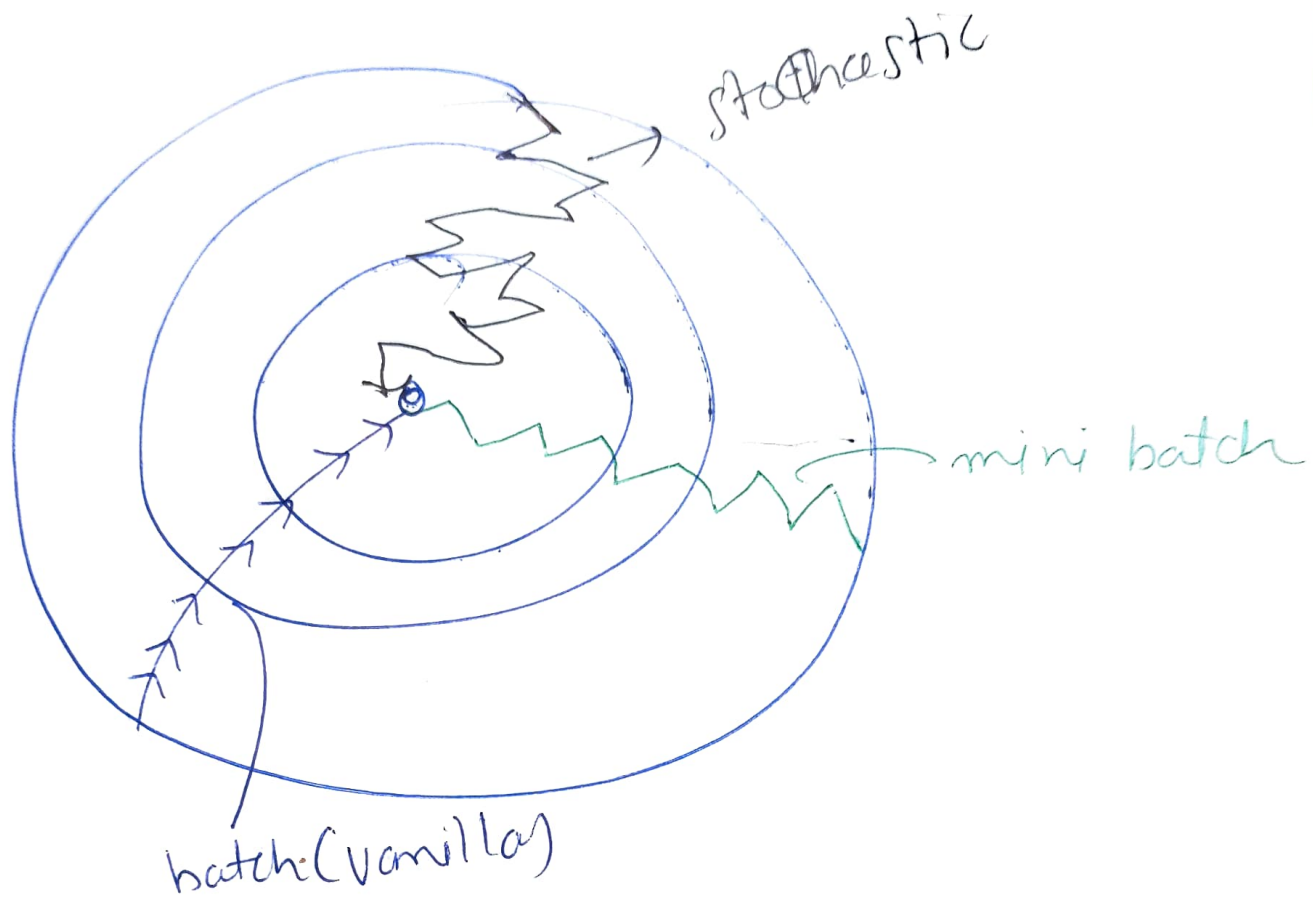
take any random point & update
the w

cover all points

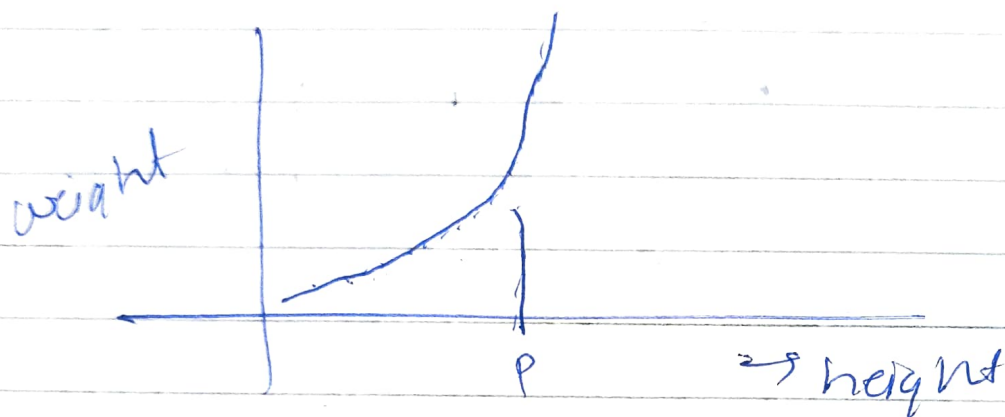
here no Σ

After completing 1 epoch (for vanilla
all points, for batch all batches, for
stochastic all points). we need to
run again next epoch, its called
iteration.

Among all the GD, stochastic GD
will have lot of fluctuations.



Height & weight



Relat betⁿ weight & height is linear upto point P, ~~but~~ i.e. during childhood after point P it is not linear, so

LR does not work here

The line is like a curve
To represent curve "quadratic"
equatⁿ needed not linear

~~Quadratic~~ mean any power above 1
 x^2, x^3, x^4
 quadratic polynomial

$$\hat{y} = w_1 f_1 + w_0 \Rightarrow \text{linear}$$

$$\hat{y} = w_1 f_1 + w_2 f_1^2 + w_0 \Rightarrow \text{quadratic}$$

~~quad~~

power at 1 $\rightarrow$ linear

2 $\rightarrow$ quadratic

> 2 $\rightarrow$ polynomial

how polynomial features look like

f_1	f_2	y
1	4	10
2	5	23
3	6	30

for polynomial we increase CG 1

f_1	f_1^2	f_2	f_2^2	y
1	1	4	16	10
2	4	5	25	23
3	9	6	36	30

as there is relation b/w f_1 , f_1^2 & f_2 f_1^2
isn't it multi-collinearity
yes, it is

Isn't it problem 9

~~It's not multi problem, its problem~~
~~for linear not for polynomial~~

It is not a problem because
 $f_1^2 = f_1 \neq f_2$

it's not a linear eqⁿ it's quadratic
so no question of multi-linearity.

$$y = m_1 x_1 + m_2 x_2 + C$$

$$Z = x_1^2$$

so y is quadratic x_1^2

Look notes here.